

LEARNING UNIT 1 · ACTIVITY 1

Retail Banking

Customer Onboarding

Build an AI agent that decides account applications end to end

PLATFORM

Microsoft Copilot Studio

DATA

SharePoint Lists

DURATION

90 minutes

01



The Problem

Why Marina Trust Bank needs this

SCENARIO

Marina Trust Bank today

Paper and re-keying

Applications arrive on paper. Staff type them into a spreadsheet by hand. Every keystroke is a chance to introduce an error.

Days, not minutes

Applications sit in a queue for 3–5 working days before anyone makes a decision. Applicants hear nothing in the meantime.

Rules in a binder

Eligibility rules are printed and kept at the branch. Different officers read them differently, so outcomes vary.

Marina Trust Bank is fictitious. All data in this lab is mock data.

What goes wrong

1

Duplicate customers

No one checks the register first, so the same person is onboarded twice under slightly different spellings

2

Inconsistent decisions

Two officers, same application, different outcome — because the rules are applied from memory

3

Missed confirmations

Emails are typed one at a time, and some are simply forgotten

4

No audit trail

When the regulator asks why an application was rejected, nobody can say



**Every decision should take twenty seconds,
follow the same rules, and leave a record.**

That is what you are going to build.

02



What You Build

The moving parts

The whole system



Response comes BEFORE the email on purpose: the applicant sees the decision even if mail delivery later fails. Never couple the user's answer to a side effect.

Four concepts

HTTP Request Trigger

A public URL that starts the flow. Any website can POST an application to it. Same idea as a webhook — the flow sits waiting until data arrives.

Agent Node

The decision maker. You give it the bank's rules in plain English and a JSON schema for its answer. It reasons over the application and returns a structured decision.

SharePoint Lists

The bank's records. Customers is the register to check against; OnboardingLog is the audit trail. Reads and writes, live.

...and the fourth

Response Action

Sends the decision back to whoever called the flow. Without it, the website gets an empty reply and can display nothing.

Why it matters

The trigger and the response are a matched pair. One opens the conversation with the website, the other closes it.

Order matters

The SharePoint lookup runs BEFORE the agent, so the agent already knows whether this person is an existing customer.

03

Inside the Agent

What the Agent node can do

Six capabilities

Instructions

The system prompt. The rules, the definitions, the output contract. Everything the agent knows about its job lives here.

Microsoft IQ

Grounds the agent in your organisation's work context — people, meetings, documents, app signals from Microsoft 365.

Tools

Systems the agent can call itself: SharePoint, Outlook, Teams, OneDrive, Excel Online, Dataverse, MCP servers, any connector.

Knowledge

Trusted reference material the agent reads from when answering, instead of relying on what the model already knows.

Request human assistance

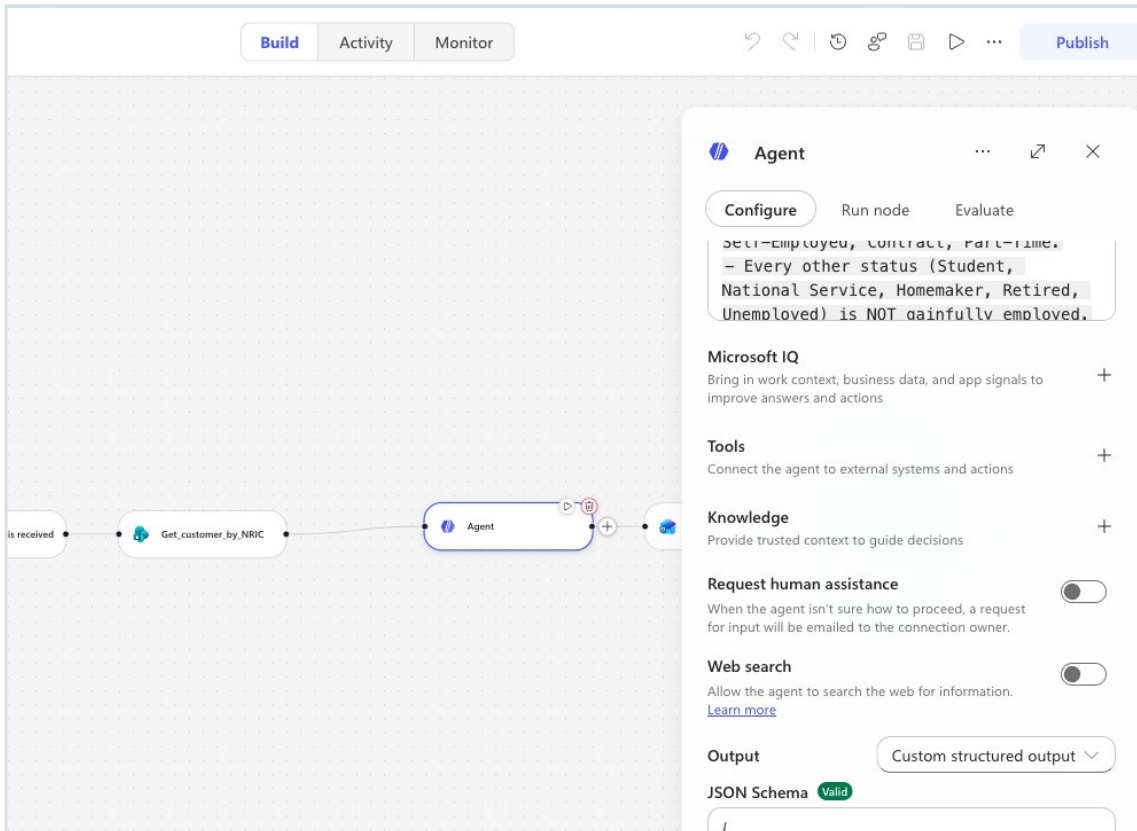
When the agent is unsure, it emails the connection owner and waits for a person to weigh in before continuing.

Web search

Lets the agent look things up on the public web. Off by default, and best left off for regulated decisions.

Activity 1 uses Instructions and Structured output only. Everything else is off — deliberately, because a bank decision should depend on the bank's own data and nothing else.

Where you find them



All six sit together at the bottom of the Agent panel:

Microsoft IQ
Tools
Knowledge
Request human assistance
Web search
Output

Activity 1 leaves both toggles off.

Structured output

Text response

The agent replies in prose. Fine for a chat assistant, useless when a website has to read the answer and render it.

Structured output

A predefined shape. The agent must fill in the fields you specified, so downstream steps can rely on them.

Custom structured output

You supply the JSON Schema. This is what Activity 1 uses — with an enum on 'decision' so only four values are possible.

A schema turns the model from something that writes an answer into something that returns data your flow can act on.

04

The Decision Logic

Six rules, in order

HOW THE AGENT DECIDES

The rules stop at the first match

STEP 1	NRIC already in the Customers register	DUPLICATE
STEP 2	Applicant is under 18 years old	REJECTED
STEP 3	PEP, foreign tax resident, or high-risk funds	REVIEW
STEP 4	Employment or income fails the product rule	REJECTED
STEP 5	Initial deposit below the product minimum	REJECTED
STEP 6	Nothing fired	APPROVED

Order is the point. An applicant who is under 18 AND a PEP is REJECTED, not REVIEW — Step 2 fires first and stops the ladder.

The product matrix

Employment requirement

Savings — none
Joint Savings — none
Student Account — must be Student
Current — gainfully employed
Fixed Deposit — income $\geq$ SGD 30,000
Multi-Currency — $\geq$ SGD 60,000 + employed

Minimum initial deposit

Savings — SGD 500
Joint Savings — SGD 1,000
Student Account — SGD 0
Current — SGD 3,000
Fixed Deposit — SGD 10,000
Multi-Currency — SGD 5,000

The agent judges the applicant only against the account type they actually asked for.

What the agent returns

Every decision comes back in this exact shape

```
{  
  "applicationId": "APP-20260801085935",  
  "decision": "APPROVED",  
  "reason": "The application has been reviewed and all eligibility  
            criteria for a Savings account have been met.",  
  "riskFlags": []  
}
```

A JSON schema with an enum on 'decision' stops the model inventing values the website cannot render.

05



Build It

Five nodes, in order

The finished flow



Trigger → SharePoint lookup → Agent → Response → Email

CONFIGURATION

Node 1 — HTTP request trigger

When a HTTP request is received ... X

Trigger type

When a HTTP request is received
This trigger fires when a HTTP request is received

Trigger the workflow via HTTP request. Configure the allowed methods and expected request schema.

Allowed HTTP method *

POST

Who can trigger the flow?

Anyone (no authentication)

Choose who can invoke this HTTP endpoint. Defaults to users in your tenant.

Relative path

/api/trigger

Optional path appended to the base webhook URL

Request body JSON schema

```
{
  "type": "object",
  "properties": {
    "fullName": {
      "type": "string"
    }
  }
}
```

METHOD
POST

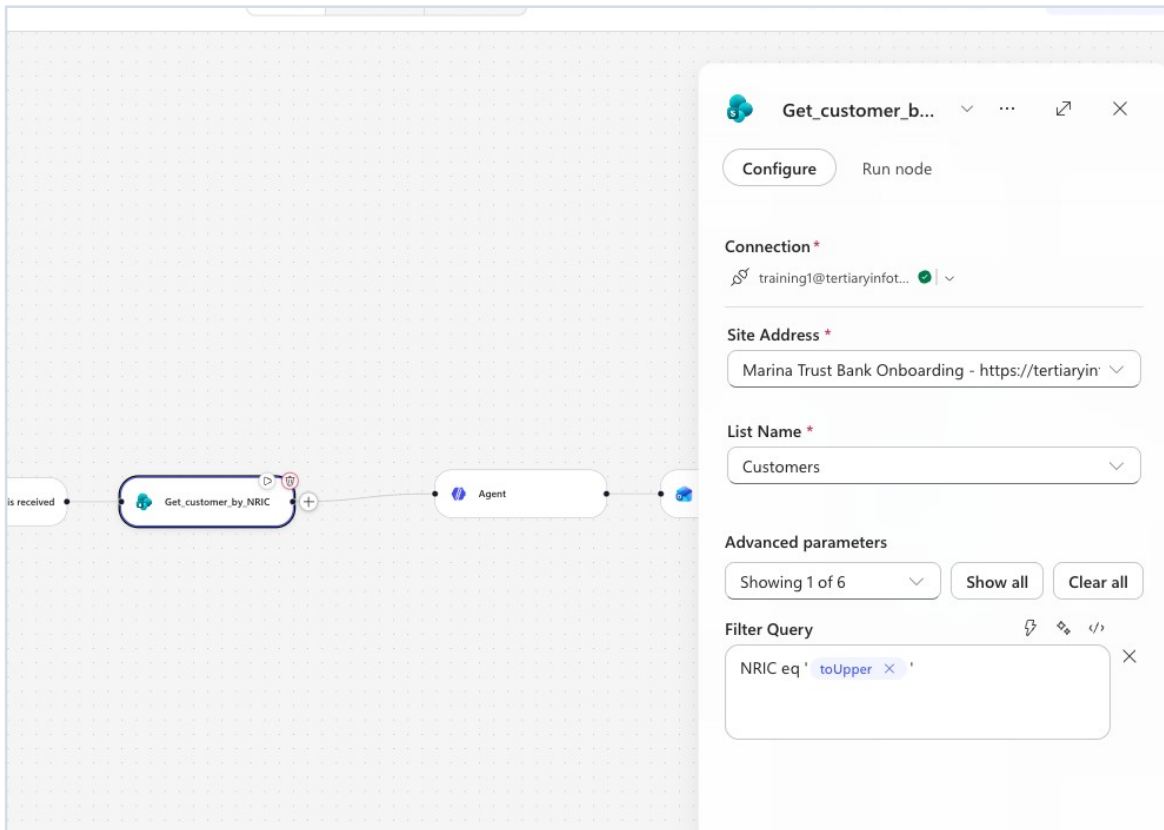
WHO CAN TRIGGER
Anyone (no authentication)

RELATIVE PATH
Leave blank — a value here
breaks Publish

The POST URL appears
only after you save.

CONFIGURATION

Node 2 — SharePoint lookup



SITE

Marina Trust Bank
Onboarding

LIST

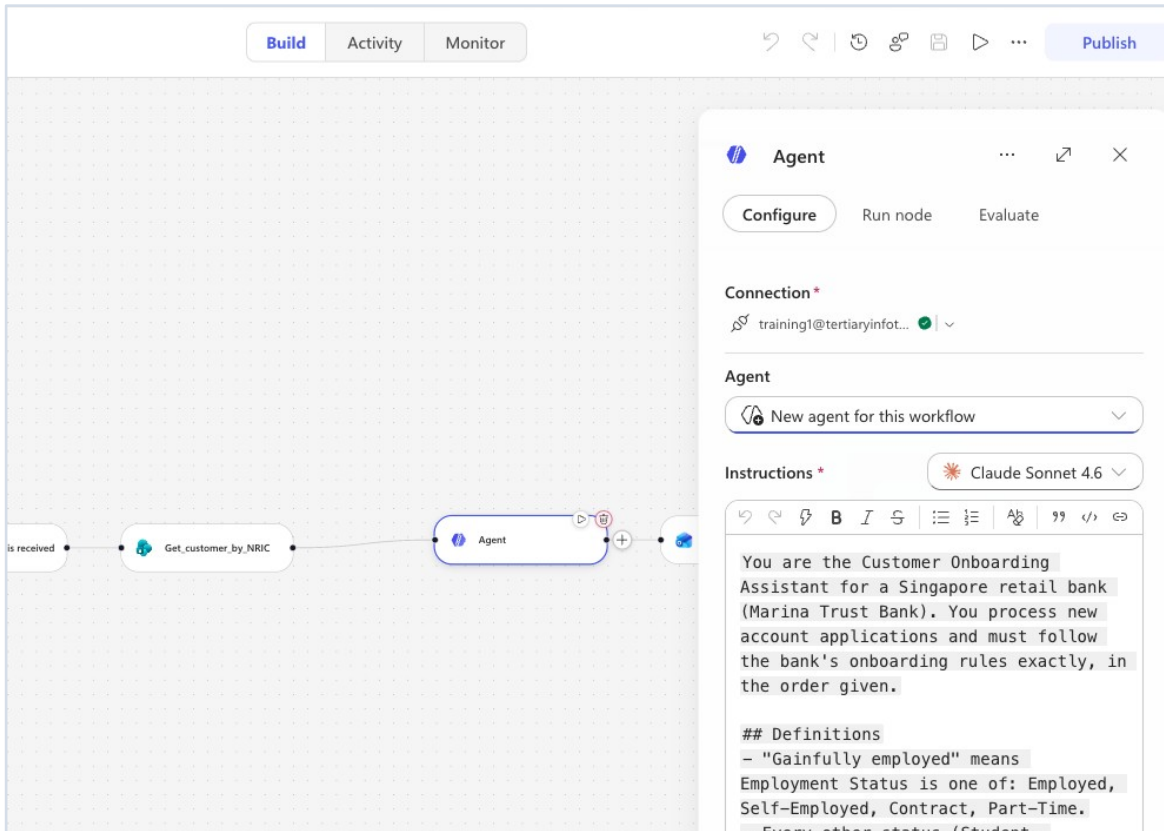
Customers

FILTER QUERY

NRIC eq '<toUpper>'

toUpper matters:
lowercase input must
still match the register.

Node 3 — the Agent



INSTRUCTIONS

The six rules, plus the application data at the bottom

OUTPUT

Custom structured output

There is no separate input field — the application goes in the Instructions.

Node 4 — the Response

The screenshot shows a workflow editor with a sequence of nodes: 'Agent', 'Send an email', and 'Response'. The 'Response' node is selected, and its configuration panel is open on the right. The panel is titled 'Response' and includes a description: 'Send a response to an HTTP request.' The 'Status Code' field is set to '200'. Under 'Advanced parameters', there are buttons for 'Showing 2 of 2', 'Show all', and 'Clear all'. The 'Headers' section shows a single header 'Content-Type' with a value of 'application/json'. There is an 'Add pair' button. The 'Body' section shows a value of 'structuredOutput'.

Response

Send a response to an HTTP request.

Status Code *

200

Advanced parameters

Showing 2 of 2 Show all Clear all

Headers

Content-Type application/json

+ Add pair

Body

structuredOutput

STATUS CODE

200

HEADERS

Content-Type:
application/json

BODY

The agent's
structuredOutput

This is what the
website renders.

Six things that will catch you

Publish, do not just save

The HTTP endpoint always serves the published version. Check the Version history panel if an edit seems to do nothing.

Relative path must be empty

Typing a path there makes Publish fail with an 'inputs.relativePath is not valid' error.

The trigger validates types strictly

Numbers must arrive as numbers, booleans as booleans. HTML forms send strings — the website has to cast them first.

Rich-text fields eat expressions

The email Body and the agent Instructions need the token picker or code mode. Pasted `@{...}` stays as dead text.

The email To field needs the picker

A typed expression arrives with a trailing newline and Outlook rejects the address. Insert the Email token with the picker instead.

HTTP 200 does not mean it worked

Once Response runs first, later actions fail invisibly. Judge success from the Activity tab, not the status code.

06



The Data

SharePoint as the bank's records

Two lists

Customers

The register to check against

NRIC · Full Name · Date of Birth

Email · Account Type

Annual Income · Onboarded On

Seeded with 5 customers

OnboardingLog

The audit trail

Timestamp · Application ID

NRIC · Account Type

Decision · Reason

One row per decision

Title is mandatory on every SharePoint item and cannot be removed. The flow sets it to the NRIC.

The Customers register

SharePoint

Search this list

MT Marina Trust Bank Onboarding

Private group Following 1 member

Discover Publish Build OneDrive

Home

Conversations

Documents

Notebook

Pages

Site contents

Recycle bin

Edit

Return to classic SharePoint

Keep Lists fully optimized. Select Allow to connect to local devices to keep things speedy. [Learn more](#) Allow X

+ New Edit in grid view Undo Share Copy link Copilot Export Forms Details

Customers ☆

Title	NRIC	Full Name	Date of Birth	Email	Account Type	Ann
S8412345D	S8412345D	TAN WEI MING	7/2/1984	tanweiming@ex...	Savings	
S9078234B	S9078234B	NURUL AISYAH BINTE RAHMAN	11/15/1990	nurul.aisyah@e...	Current	
S7623451A	S7623451A	RAJESH KUMAR	4/8/1976	rajesh.kumar@e...	Fixed Deposit	
T0145678C	T0145678C	CHLOE LIM HUI LING	9/23/2001	chloe.lim@exa...	Student Account	
S6534129E	S6534129E	GOH BEE CHOO	2/11/1965	goh.beechoo@...	Savings	
S8823417F	S8823417F	LIM JIA HAO	5/19/1988	lim.jiahao@exa...	Savings	
S9234871G	S9234871G	PRIYA MENON	8/30/1992	priya.menon@e...	Current	
S7412903H	S7412903H	ABDUL RAHIM BIN OSMAN	12/5/1974	abdul.rahim@e...	Fixed Deposit	
T0298345J	T0298345J	MARCUS TEO WEI JIE	3/17/2002	marcus.teo@ex...	Student Account	

All Items Add view

This is what the duplicate check queries on every application

07



Run It

The applicant's experience

The application form



[How it works](#) [Accounts](#) [Open an account](#)

Training environment. Marina Trust Bank is a fictitious bank used for this lab. Do not enter a real NRIC, address or email belonging to anyone other than yourself.

PERSONAL BANKING

Open your account in minutes, not days.

Submit one form. Our onboarding assistant checks your details against our customer records, runs the required KYC screening, and gives you a decision on the spot — no branch visit, no paperwork queue.

[Start your application](#)

Takes about 3 minutes · Decision shown immediately

Application received	✓
Existing customer check	✓
KYC / AML screening	✓
Eligibility assessment	✓
Decision & confirmation email	✓

How your application is processed

Every application runs through the same four checks, in the same order, every time. The checks stop at the first one that fails — so you always know exactly which criterion decided the outcome.

1 Existing customer check

2 Age & identity

Paste your HTTP POST URL into Lab configuration, then submit an application

RESULT

An approved application

MT

Marina Trust Bank
SINGAPORE

How it worksAccountsOpen an account

TRAINER DEMO DATA

TC1 - Happy path (expect APPROVED)

Autofills the form so you can demonstrate a decision path quickly. Remember to set your own email address.

✓

Application approved
The application has been reviewed and all eligibility criteria for a Savings account have been met. We are pleased to approve the account opening for Mr Marcus Ng Jun Hao.

ReferenceAPP-20260801092846

Confirmation emailNot sent

Submit another application

Marina Trust Bank — a fictitious bank created for the NTUC LearningHub course Building AI Agents for Work Automation. No real banking service is offered and no real accounts are opened.

LU1 Activity 1 - Powered by a Copilot Studio agent flow

Powered by Tertiary Infotech Academy Pte Ltd

WHAT HAPPENED

1. Form posted the application
2. SharePoint found no matching NRIC
3. Agent applied the rules, nothing fired
4. Decision returned and rendered here

RESULT

An existing customer

MT Marina Trust Bank
SINGAPORE

How it worksAccountsOpen an account

TRAINER DEMO DATA

TC2 - Existing customer (expect DUPLICATE)

Autofills the form so you can demonstrate a decision path quickly. Remember to set your own email address.

!

You are already a customer

This application cannot be processed as the applicant is already an existing customer in our records. No new account has been opened.

ReferenceAPP-20260801093001

Confirmation emailNot sent

Submit another application

Marina Trust Bank — a fictitious bank created for the NTUC LearningHub course Building AI Agents for Work Automation. No real banking service is offered and no real accounts are opened.

LU1 Activity 1 - Powered by a Copilot Studio agent flow

Powered by Tertiary Infotech Academy Pte Ltd

WHAT HAPPENED

1. Form posted the application
2. SharePoint FOUND the NRIC in Customers
3. Step 1 fired and stopped the ladder
4. No account opened

VERIFICATION

Test every rule

TEST CASE	INPUT	EXPECTED
TC1 Happy path	New NRIC, Savings, SGD 1,000	APPROVED
TC2 Existing customer	S8412345D	DUPLICATE
TC5 Messy casing	s8412345d lowercase	DUPLICATE
TC6 Politically exposed	PEP = Yes	REVIEW + PEP
TC7 Under 18	Date of birth 2010	REJECTED + MINOR
TC8 Deposit too low	Savings, SGD 200	REJECTED

TC5 is the one to demonstrate: same NRIC as TC2 but lowercase. Without normalisation it would be approved and create a duplicate.



**Expect about twenty seconds
per application.**

The SharePoint lookup and the model call are both network round-trips. It has not hung.

08

Step or Tool?

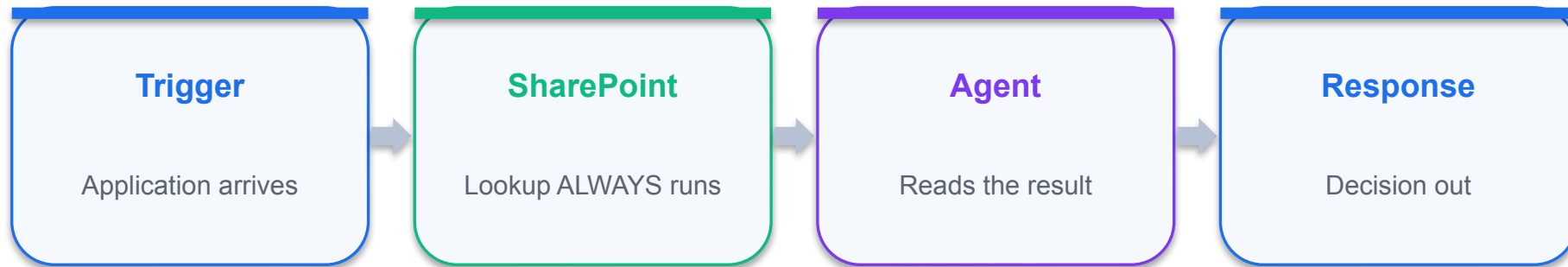
Two ways to give an agent data



"Why didn't we add SharePoint as a Tool in the agent?"

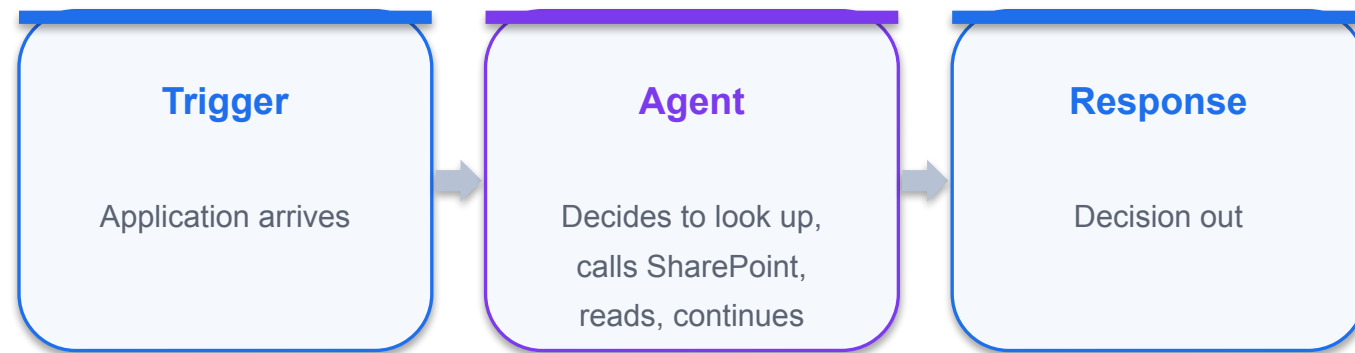
A fair question — and the answer is the most important idea in this learning unit.

Activity 1 — SharePoint as a step



You decided the order at design time. The lookup cannot be skipped, because it is not the agent's choice to make.

Activity 1b — SharePoint as a tool



The agent decides at run time whether and when to check. The lookup is inside its reasoning, not on the canvas.

The tradeoff in full

	SharePoint as a STEP	SharePoint as a TOOL
Who decides	You, at design time	Agent, at run time
Lookup guaranteed	Yes — structural	No — instructed
Model calls per run	1	2 or more
Cost	Lower	Higher
Visible on canvas	Yes	No — inside the agent
Auditability	Read the run history	Read the reasoning
Handles the unexpected	Only what you built	Can adapt

Neither column is the right answer. The question is always: does this task need guarantees, or does it need judgement?

Why Activity 1 uses a step

1

Rule 1 says "always first"

As a step that is structurally guaranteed. As a tool you are trusting the model to always call it

2

The failure is expensive

An agent that skips the duplicate check opens a second account for an existing customer — the exact thing this automation exists to prevent

3

A regulator will ask

"Show me that you checked." A step appears in every run history. A tool call is buried in the agent's reasoning

4

Credits are finite

Tool use means reason, call, reason again. In a Developer environment with a class running tests, that adds up

This is not a limitation of Copilot Studio. It is a design decision, and it is the kind you will make on every real project.

When to use which

Use a STEP when

- The check is mandatory
- Order is a compliance requirement
- Every run must behave the same
- You need it provable, not likely
- Volume makes cost per run matter

Use a TOOL when

- The path is not known in advance
- Some lookups are only sometimes needed
- The agent may need to call it repeatedly
- Flexibility beats predictability
- The task genuinely needs judgement

Most production systems use both — steps for what must happen, tools for what requires judgement.



Build it both ways. Run them side by side.

Activity 1b is the same decision with SharePoint as a tool. The decisions match; everything else differs.

09



Going Further

Where this goes next

Extend the lab

1

Write the customer record

Add Create item on the APPROVED branch, so an approved applicant joins the register — resubmit and it returns DUPLICATE

2

Write the audit log

Add Create item to OnboardingLog for every decision, whatever the outcome

3

Human review for REVIEW cases

Route PEP and high-risk applications to an approval step instead of straight to the applicant

4

Compare with the n8n build

Same scenario, same rules, different platform — see what each makes easy and what each makes hard

Full step-by-step instructions, every expression and every prompt: see LAB-GUIDE.md in the lab folder.



Questions?

LAB-GUIDE.md has the complete build, expression by expression.